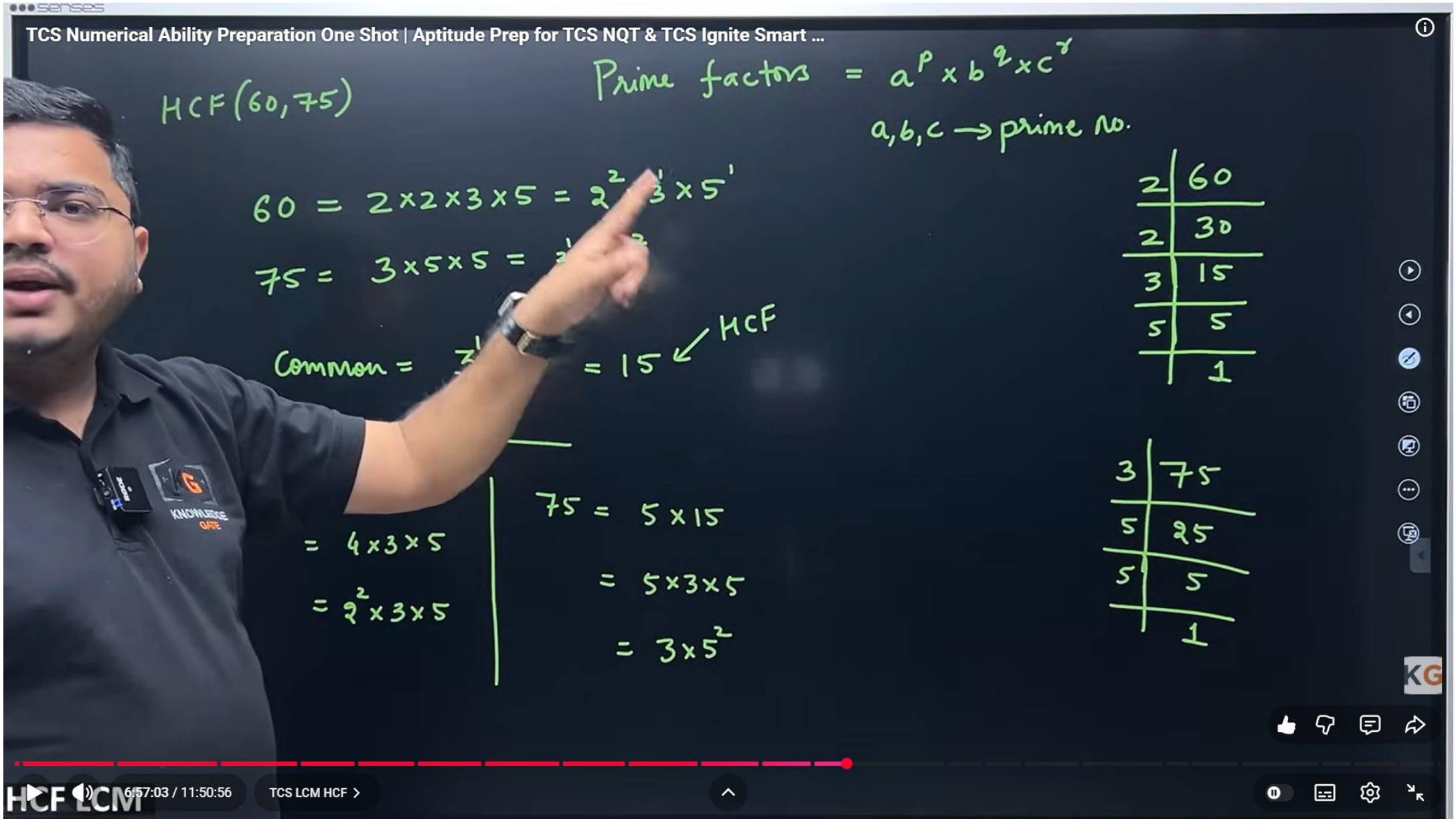
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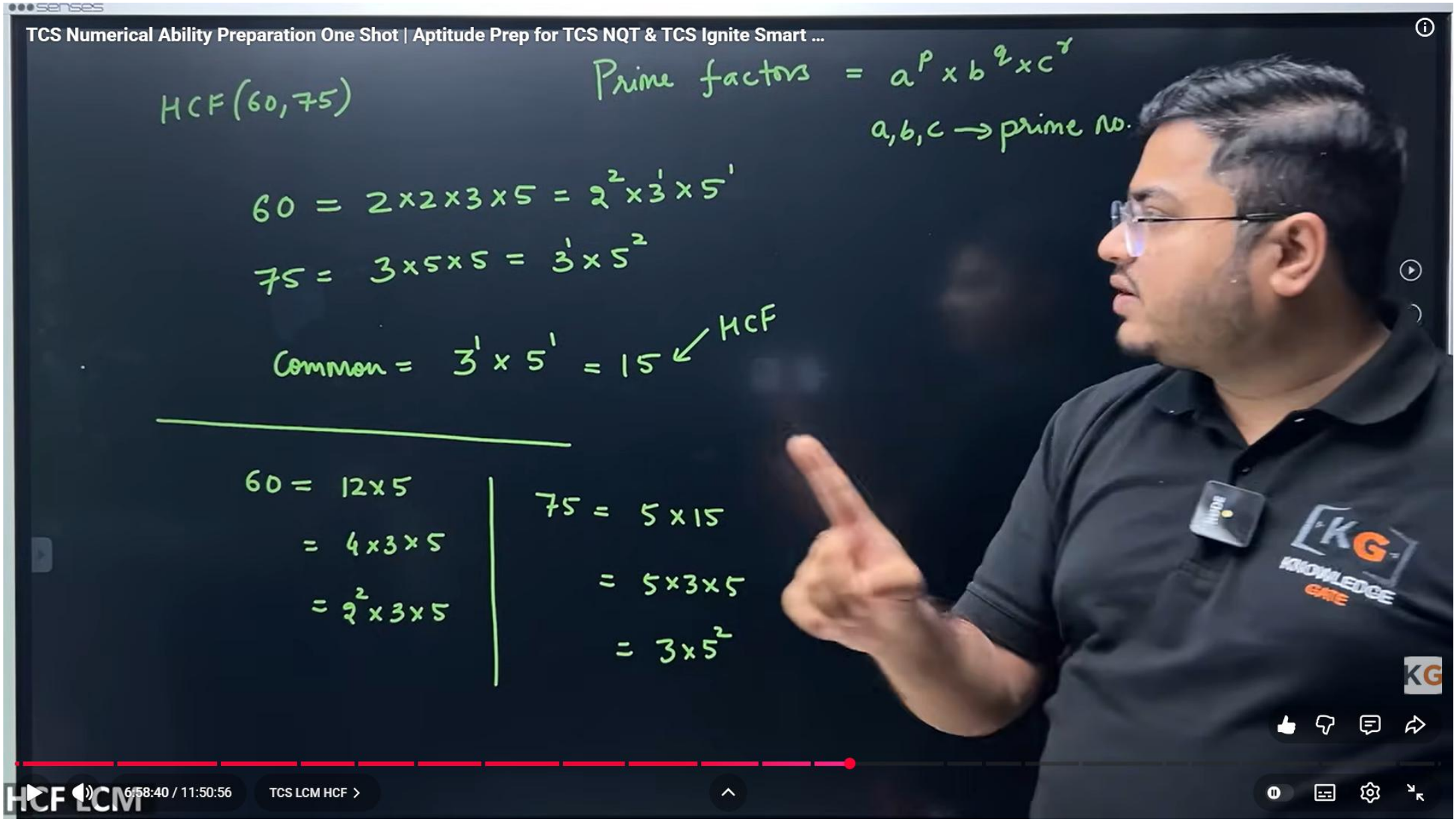
HCF(60, 75)
Prime factors = $a^p \times b^q \times c^r$
a, b, c → prime no.
60 = 2×2×3×5 = $2^2 \times 3^1 \times 5^1$
75 = 3×5×5 = $3^1 \times 5^2$
Common = $3^1 \times 5^1$ = 15 ← HCF
= 4×3×5
= $2^2 \times 3 \times 5$
75 = 5×15
= 5×3×5
= 3×5^2
2 | 60
2 | 30
3 | 15
5 | 5
1
3 | 75
5 | 25
5 | 5
1
HCF LCM
6:57:03 / 11:50:56
TCS LCM HCF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HCF(60, 75)
Prime factors = $a^p \times b^q \times c^r$
a, b, c → prime no.
$60 = 2 \times 2 \times 3 \times 5 = 2^2 \times 3^1 \times 5^1$
$75 = 3 \times 5 \times 5 = 3^1 \times 5^2$
Common = $3^1 \times 5^1 = 15$ HCF
$60 = 12 \times 5$
$= 4 \times 3 \times 5$
$= 2^2 \times 3 \times 5$
$75 = 5 \times 15$
$= 5 \times 3 \times 5$
$= 3 \times 5^2$
KG
KNOWLEDGE GATE
6:58:40 / 11:50:56
TCS LCM HCF
HCF LCM

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HCF by Prime Factorization

Follow the below-given steps to find the hcf of numbers using prime factorization method.

Step 1: Write each number as a product of its prime factors. This method is called here prime factorization.

Step 2: Now list the common factors of both the numbers

Step 3: The product of all common prime factors is the HCF (use the lower power of each common factor)

Let us understand with the help of examples.

Example 1: Evaluate the HCF of 60 and 75.

Solution:

Write each number as a product of its prime factors.

$2^2 \times 3 \times 5 = 60$

$3 \times 5^2 = 75$

The product of all common prime factors is the HCF.

The common prime factors in this example are 3 & 5.

The lowest power of 3 is 3 and 5 is 5.

So, HCF = 3 x 5 = 15

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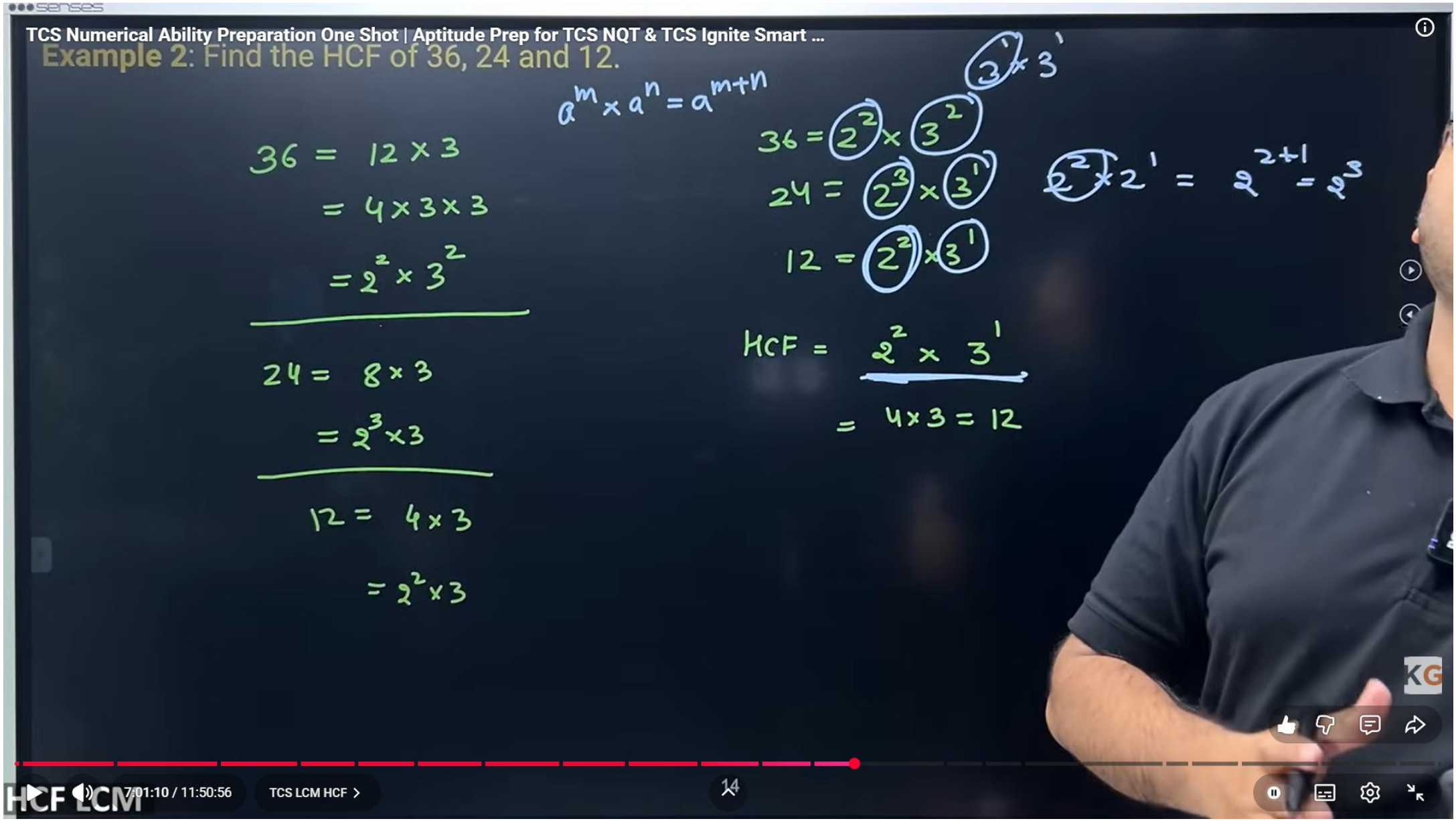
TCS LCM HCF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Example 2: Find the HCF of 36, 24 and 12.
$a^m \times a^n = a^{m+n}$
$36 = 12 \times 3$
$= 4 \times 3 \times 3$
$= 2^2 \times 3^2$
$24 = 8 \times 3$
$= 2^3 \times 3$
$12 = 4 \times 3$
$= 2^2 \times 3$
$36 = 2^2 \times 3^2$
$24 = 2^3 \times 3^1$
$12 = 2^2 \times 3^1$
$3^1 \times 3^1$
$2^2 \times 2^1 = 2^{2+1} = 2^3$
HCF = $2^2 \times 3^1$
$= 4 \times 3 = 12$
HCF LCM
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Example 2: Find the HCF of 36, 24 and 12.

$$HCF(36, 24, 12) = HCF(a, 12)$$

$$\boxed{HCF(a, a, a, a \ldots) = a}$$

$$HCF(36, 24) = a$$

$$a, b, c, d$$
$$\alpha, c, d$$
$$\beta, d$$

$$\Rightarrow LCM(a, b, c) = LCM(z, c)$$

$$z = LCM(a, b)$$

$$36 = 2^2 \times 3^2$$

$$24 = 2^3 \times 3^1$$

$$HCF = 2^2 \times 3^1 = 12$$

$$HCF(36, 24, 12) = HCF(12, 12) = 12$$

$$12 = 2^2 \times 3^1$$

$$12 = 2^2 \times 3^1$$

$$HCF = 2^2 \times 3^1 = 12$$

HCF LCM
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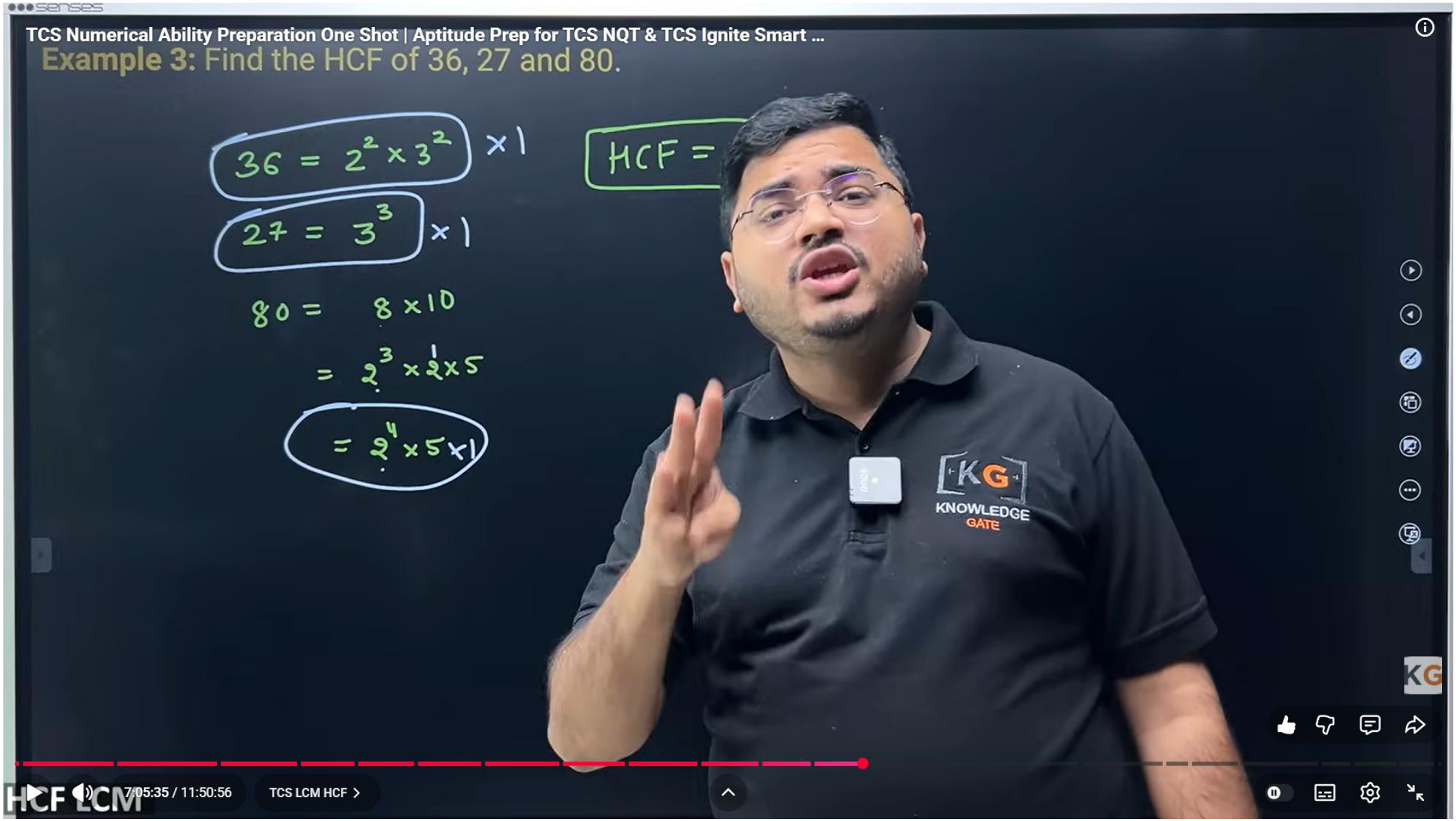
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Example 3: Find the HCF of 36, 27 and 80.
36 = 2^2 x 3^2 x 1
27 = 3^3 x 1
80 = 8 x 10
= 2^3 x 2^1 x 5
= 2^4 x 5 x 1
HCF =
KNOWLEDGE GATE
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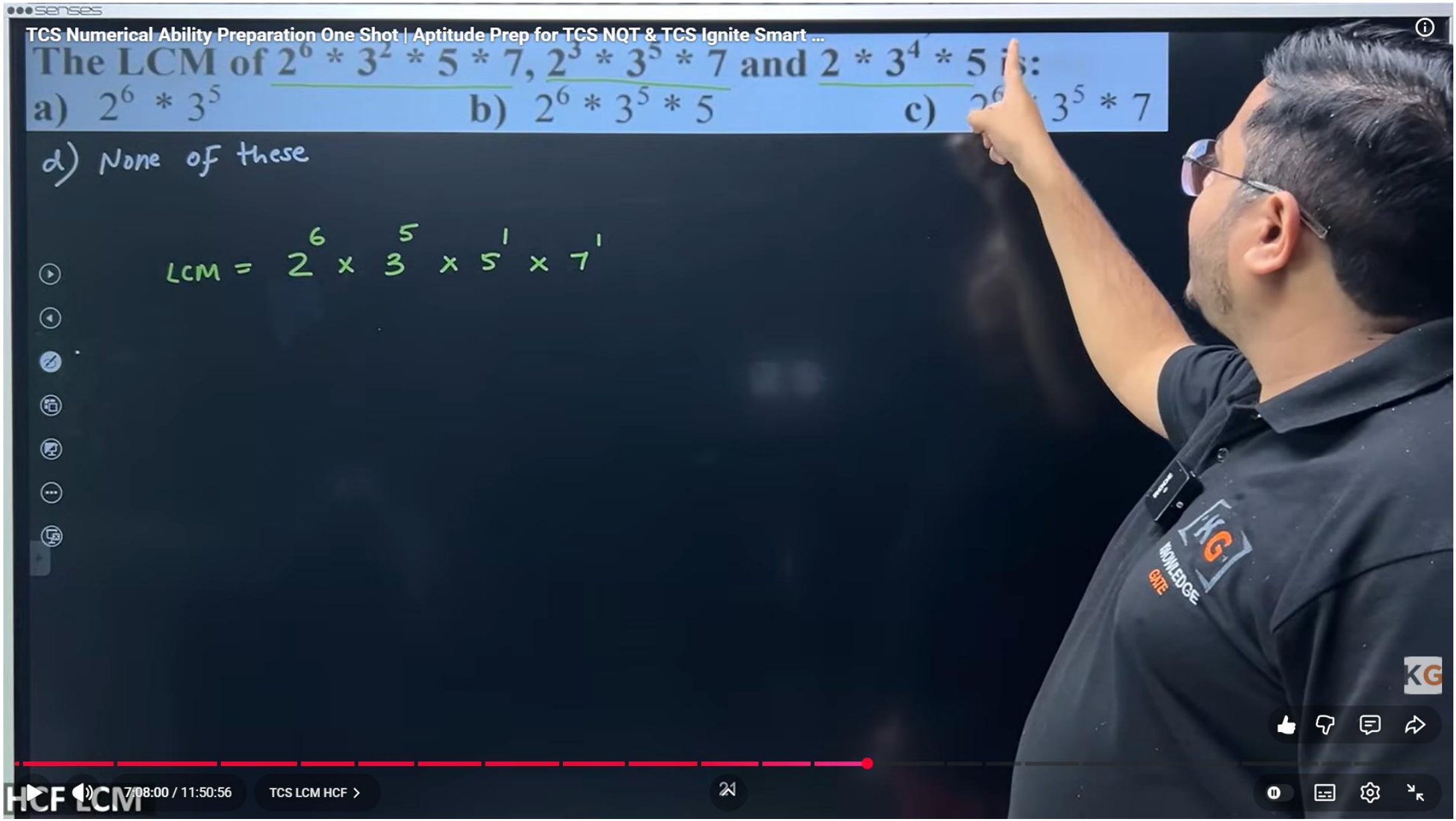
TCS LCM HCF
HCF LCM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The LCM of $2^6 * 3^2 * 5 * 7$, $2^3 * 3^5 * 7$ and $2 * 3^4 * 5$ is:
a) $2^6 * 3^5$
b) $2^6 * 3^5 * 5$
c) $3^5 * 7$
d) None of these
LCM = $2^6 \times 3^5 \times 5^1 \times 7^1$
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TCS LCM HCF
HCF LCM

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Q: If HCF of 189 and 297 is 27, find their LCM.

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HCF * LCM = Product of two numbers

$27 * LCM = 189 * 297$

$LCM = \frac{189 * 297}{27}$

$= \frac{189 * \cancel{297}^{11}}{\cancel{27}}$

$= 189 * 11$

$= 2079$

$$\begin{array}{r} 270 \\ 27 \\ \hline 297 \end{array}$$

$$\begin{array}{r} 1890 \\ 189 \\ \hline 2079 \end{array}$$

HCF LCM
7:13:09 / 11:50:56
TCS LCM HCF >
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HCF (770, 1430, 1760)

Applicable for Large Numbers

$HCF(770, 1430, 1760) = a$

$HCF(\underbrace{77, 143}_{11}, 176) = \frac{a}{10}$

z

$HCF(z, 176)$

$\boxed{z = HCF(77, 143)}$

$$\begin{array}{r} 1 \\ 77 \overline{)143} \\ -77 \\ \hline 66 \end{array}$$

$$\begin{array}{r} 1 \\ 66 \overline{)77} \\ -66 \\ \hline 11 \end{array}$$

$$\begin{array}{r} 6 \\ 11 \overline{)66} \\ -66 \\ \hline \times\times \end{array}$$

$HCF(77, 143) = 11$

$HCF(11, 176) = 11$

$$\begin{array}{r} 16 \\ 11 \overline{)176} \\ -11 \\ \hline 66 \\ -66 \\ \hline \times\times \end{array}$$

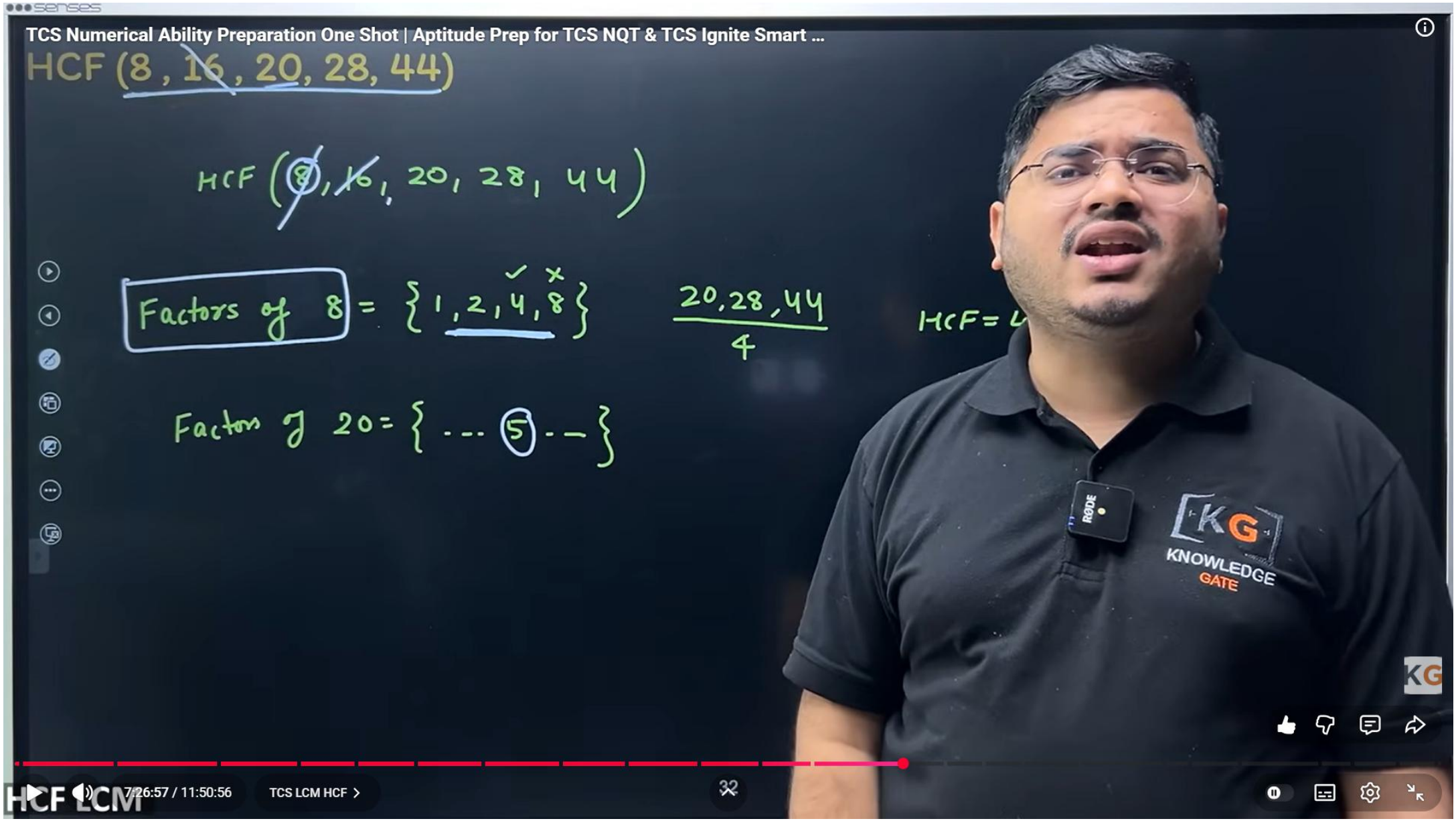
$HCF(11, 176) = 11$

$\therefore HCF(77, 143, 176) = 11$

$\frac{a}{10} = 11$

$a = 11 \times 10 = 110$

HCF LCM
7:21:56 / 11:50:56
TCS LCM HCF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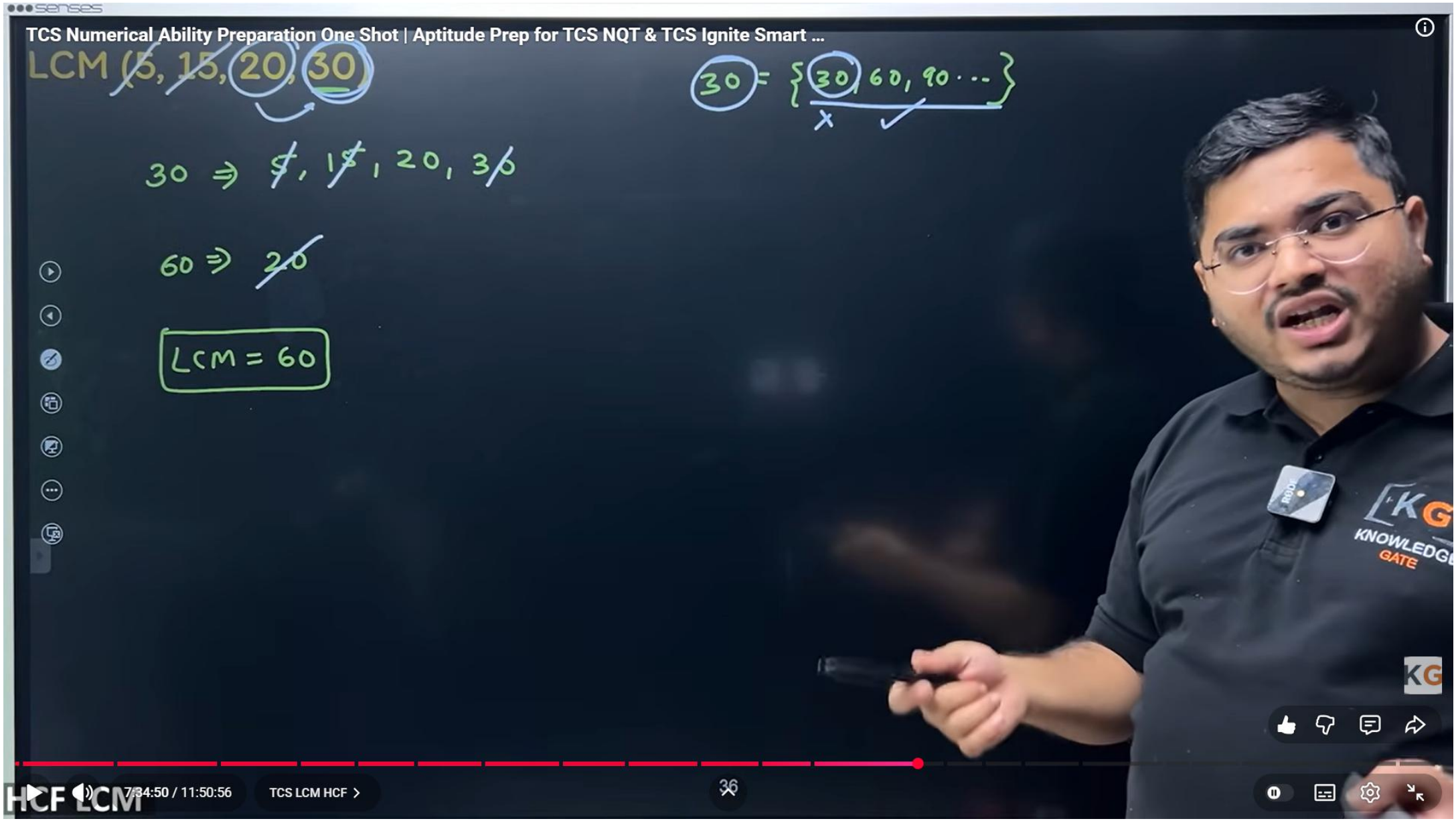
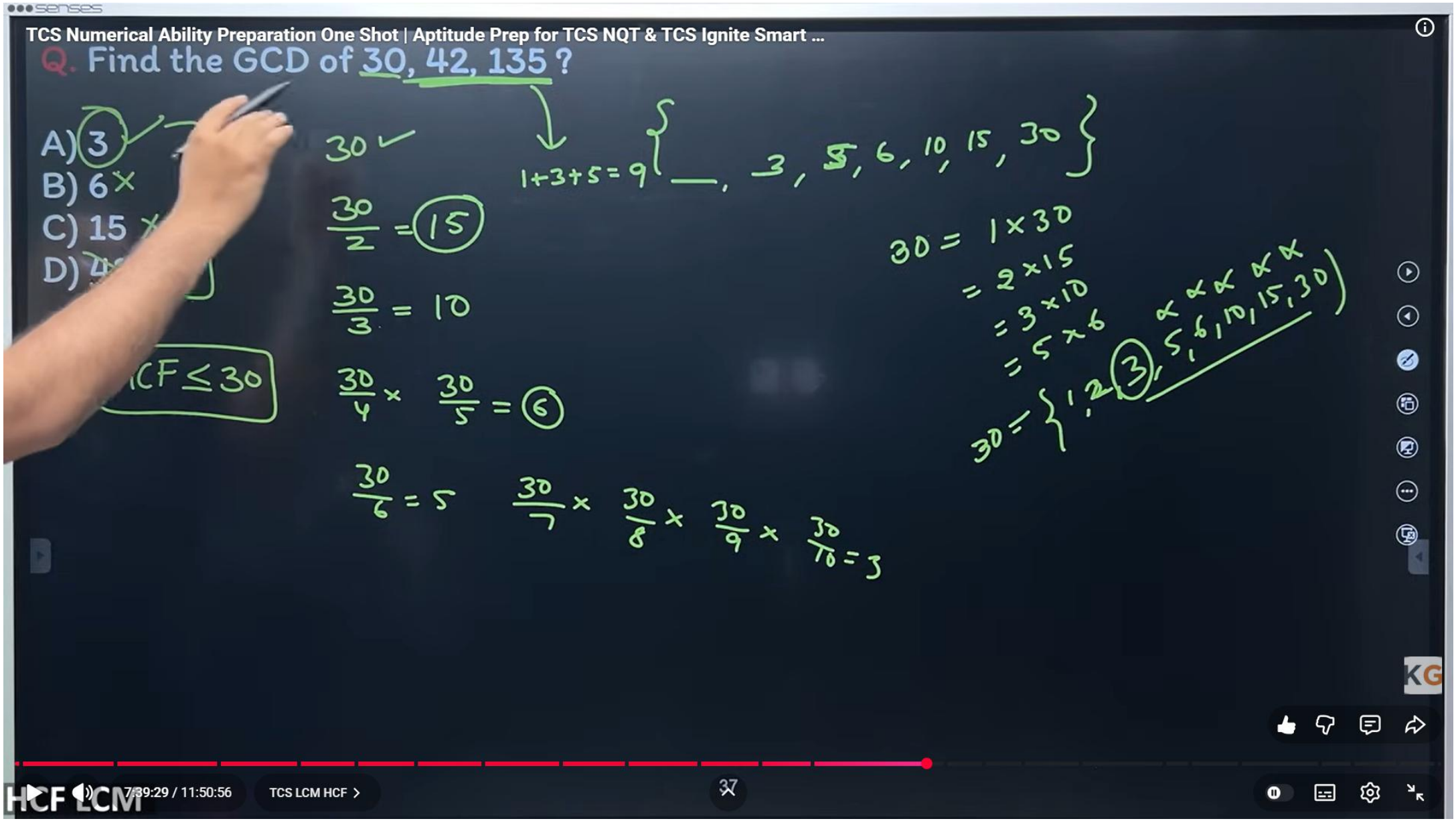
HCF (8 , 16 , 20, 28, 44)
HCF (8, 16, 20, 28, 44)
Factors of 8 = {1, 2, 4, 8}
20, 28, 44
4
HCF = 4
Factors of 20 = {... 5 ...}
KNOWLEDGE
GATE
7:26:57 / 11:50:56
TCS LCM HCF
HCF LCM


TCS Numerical Ability Preparation One Shot | Aptitude Prep for TCS NQT & TCS Ignite Smart ...
LCM (5, 15, 20, 30)
30 = {30, 60, 90 ...}
30 ⇒ 5, 15, 20, 30
60 ⇒ 20
LCM = 60
KNOWLEDGE GATE
7:34:50 / 11:50:56
TCS LCM HCF
36
HCF LCM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Q. Find the GCD of 30, 42, 135 ?
A) 3
B) 6
C) 15
D) 4
HCF ≤ 30
30
30/2 = 15
30/3 = 10
30/4 × 30/5 = 6
30/6 = 5
30/7 × 30/8 × 30/9 × 30/10 = 3
1+3+5=9
{ __, 3, 5, 6, 10, 15, 30 }
30 = 1×30
= 2×15
= 3×10
= 5×6
30 = { 1, 2, 3, 5, 6, 10, 15, 30 }
HCF LCM
7:39:29 / 11:50:56
TCS LCM HCF
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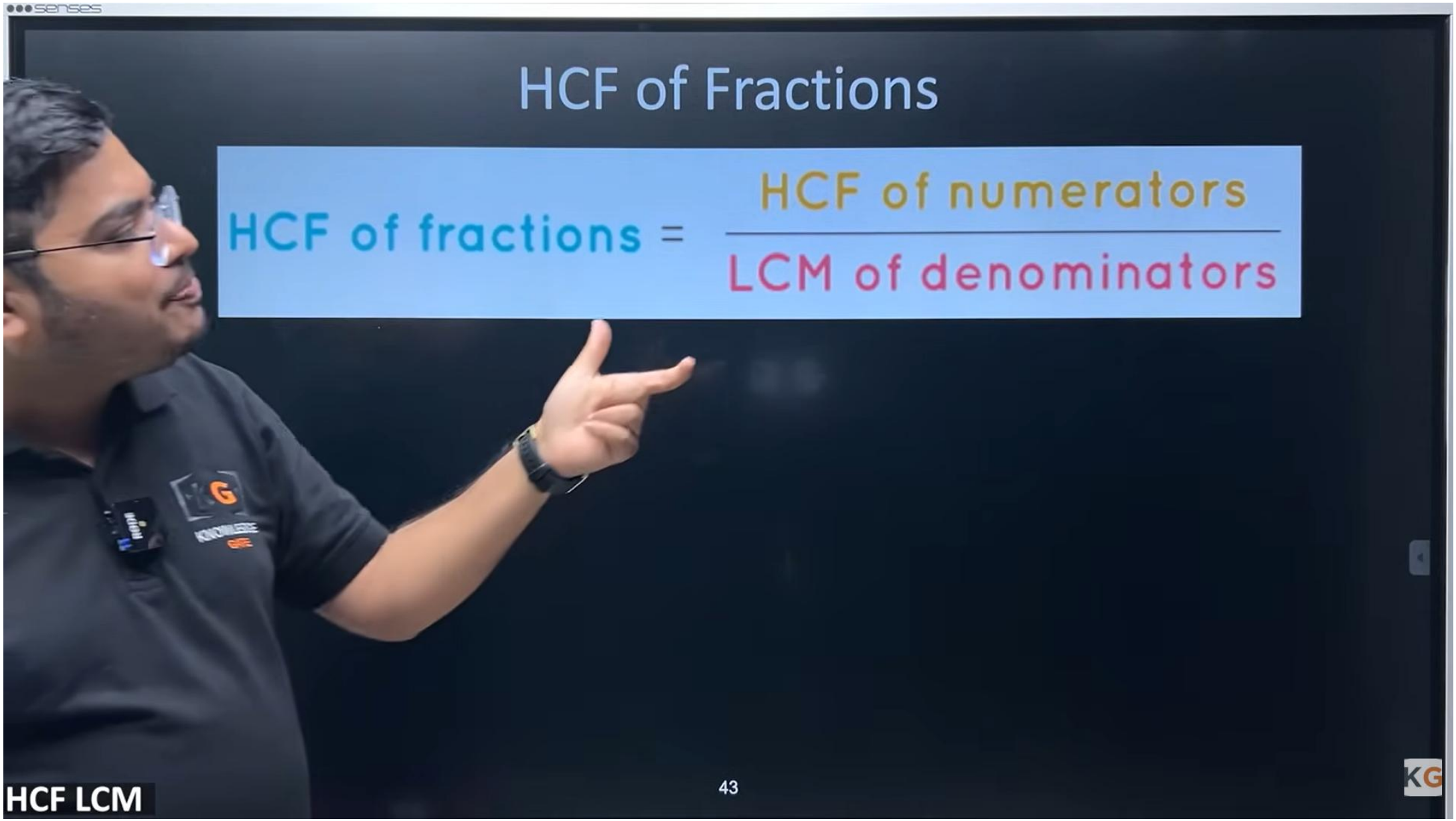
HCF of Fractions
HCF of fractions = HCF of numerators / LCM of denominators
HCF LCM
43